

Before You Start

Three reminders. Symbolic algebra, two trig identities, and what AG means.

Symbolic algebra

no specific numbers anywhere – just letters.

Multiply and simplify with letters.

Don't reach for the calculator until the question asks.

if r is symbolic, r^2 is just r^2 – leave it.

Useful trig identities

$$\sin(2\theta) = 2 \sin \theta \cos \theta.$$

$$\cos(2\theta) = 1 - 2 \sin^2 \theta = 2 \cos^2 \theta - 1 = \cos^2 \theta - \sin^2 \theta.$$

$$\sin^2 \theta + \cos^2 \theta = 1.$$

you'll need these to match a target form.

Especially $\sin(2\theta) = 2 \sin \theta \cos \theta$.

AG = 'Answer Given'

the question gives you the target.

Your job: show every algebraic step.

Examiners deduct for jumps.

If the target is, for example, $2r\theta \cos(\theta)$, your last line may use any algebraically equivalent product, such as $2\theta r \cos(\theta)$. Make the equivalence clear.

In show-that questions, prove an expression equivalent to the target. Matching its displayed form is helpful, but reordered multiplication is still correct.

What Does 'In Terms Of' Mean?

The final expression must use ONLY the variables named in the question.
No extras.

Acceptable

Question:

Find the perimeter in terms of r and θ .

Answer:

$$P = 2r + 2r\theta + 2r \sin(\theta).$$



Only r and θ -
exactly what was asked.

Not acceptable

Question:

Find the perimeter in terms of r and θ .

Answer:

$$P = 2r + 2r\theta + a + b.$$



Contains extra letters a and b .
Even if you defined them -
they're not in 'in terms of'.

Asked for	Acceptable	Not OK
in terms of r and θ	$P = 2r + 2r\theta + 2r \sin\left(\frac{\theta}{2}\right).$ 	$P = 2r + \text{arc} + \text{chord}.$ 
in terms of x and θ	$\text{Area} = 2x^2 \theta \cos^2 \theta.$ 	$\text{Area} = OA^2 \theta \cos^2 \theta.$ 
in terms of r and α	$\text{Segment area} = r^2 \left(\alpha - \frac{1}{2} \sin(2\alpha) \right).$ 	$\text{Segment area} = r^2 (\alpha - \sin \alpha).$ 

before writing the final line, scan:
every letter in the answer must be in the question's
'in terms of' list.

Show-That Strategy

Read the target carefully. Choose the formula that lands closest.
Show every step.

Read the target

Target:

Show that arc length AB
 $= 2 \times \theta \cos \theta$.

Notice the structure:
2, $\times$, θ , $\cos \theta$.
Each one comes from
somewhere in the working.

Choose the formula

Arc length = (radius) $\times$ (angle).

we need to match $2 \times \cos \theta$
as the radius and θ as
the angle. So we need to
SHOW the radius is $2 \times \cos \theta$
first.

The 4-step show-that method

1. READ the target. Note its structure (factors, signs, trig functions).
2. CHOOSE the matching formula (arc length, sector area, chord, etc.).
3. WORK in symbols, derive each piece. Match the target's exact format.
4. CONCLUDE with the target or a clearly equivalent expression. If you use a nontrivial identity or factorisation, show the equivalence explicitly. Underline or pill the result.

every algebraic step visible. If you simplify $2 \sin \theta \cos \theta$ to $\sin(2\theta)$, write the rule first, then the substitution.

The conclusion should establish the target.
Algebraically equivalent forms are valid when their equivalence is clear.

Method: Symbolic Computation

Four steps. Same as exact form, but with letters everywhere.

The 4-step method

1. SET UP the relevant formula(s) in their symbolic form.
(chord = $2r \sin\left(\frac{\theta}{2}\right)$, sector = $\left(\frac{1}{2}\right)r^2\theta$, etc.)
2. SUBSTITUTE the symbolic variables given by the question
(r for radius, x for some other length, θ for an angle).
3. SIMPLIFY using algebra and trig identities.
Combine terms. Factor when possible.
4. PRESENT the final expression in the format requested:
"in terms of r and θ ", "in exact form", or "show that ...".

keep θ in radians. Every formula in this lesson assumes θ in radians.

Common symbolic identities

Expression	Equivalent form
$2 \sin \theta \cos \theta$	$\sin(2\theta)$
$1 - \cos(2\theta)$	$2 \sin^2 \theta$
$1 + \cos(2\theta)$	$2 \cos^2 \theta$
$r^2 - r^2 \cos \theta$	$r^2(1 - \cos \theta)$
$\theta - \sin \theta$	stays as is - no simplification
$2r \sin\left(\frac{\theta}{2}\right)$	chord, half-angle form

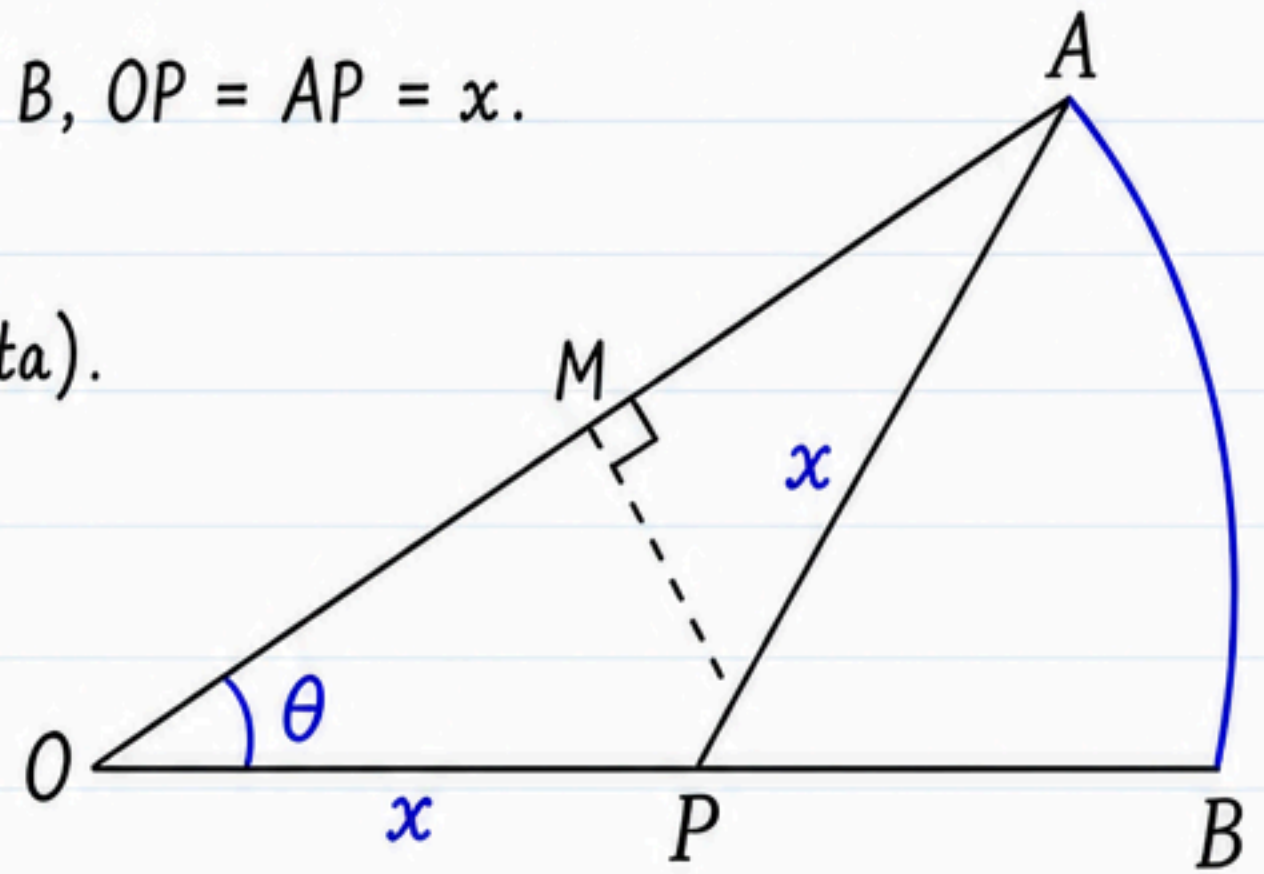
use double-angle identities ONLY when the target uses them.
Don't "expand" for its own sake.

First Worked Example - Show That Arc Length = $2x \theta \cos(\theta)$

Sector OAB. P on OB between O and B, $OP = AP = x$.

Angle AOP = θ .

Show that arc AB = $2x \theta \cos(\theta)$.



1. Triangle OAP is isosceles ($OP = AP = x$),
so angle OAP = angle AOP = θ . base angles equal.
2. In triangle OAP: angle APO = $\pi - 2\theta$. angle sum.
3. Drop a perpendicular from P to OA
at midpoint M.
Then $OM = AM = \frac{OA}{2}$. perpendicular
bisector.
4. In right triangle OPM:
 $OM = OP \cos(\theta) = x \cos(\theta)$. cos = adj / hyp.
5. So $OA = 2OM = 2x \cos(\theta)$. radius of sector.
6. Arc AB = OA $\times$ angle AOB. arc = radius $\times$ angle.
7. angle AOB = θ
because P lies on OB. straight line.
8. Arc AB = $(2x \cos(\theta)) \theta$
= $2x \theta \cos(\theta)$. AG. AG.

$$\text{arc AB} = 2x \theta \cos(\theta) \quad \text{AG}$$

Show enough valid working to earn the question-specific method and accuracy marks. There is no fixed one-mark deduction per omitted line.

the radius of the sector = $2x \cos(\theta)$.

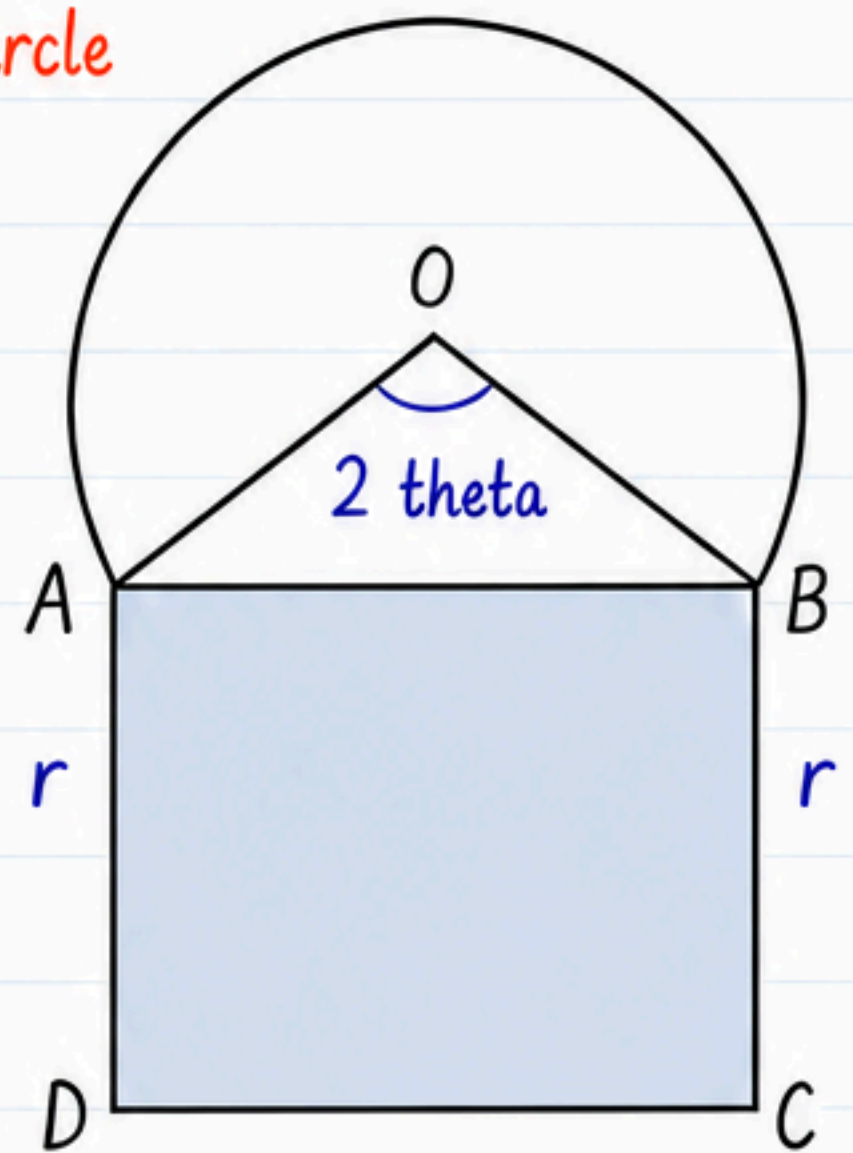
Once you have the radius, the arc = radius $\times$ θ is easy.

Worked Examples - Perimeter in r and θ

Two examples expressing perimeter symbolically.

Worked Example 2 - rectangle attached to circle

In the diagram, O is the centre of the circle of radius r . The rectangle $ABCD$ has $AD = BC = r$. Angle $AOB = 2\theta$. Find perimeter of the shaded region in terms of r and θ .



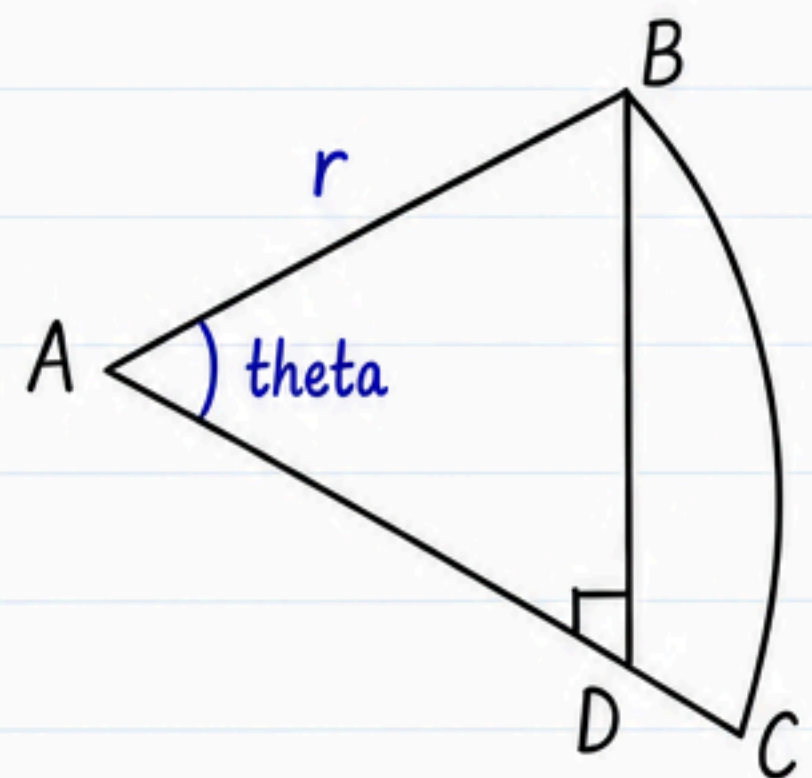
1. Chord $AB = 2r \sin \theta$.
2. $AD + BC = 2r$ and $CD = AB = 2r \sin \theta$.
3. $P = AD + DC + CB + BA$
 $= r + 2r \sin \theta + r + 2r \sin \theta$.
4. $P = 2r + 4r \sin \theta$
 $= 2r(1 + 2 \sin \theta)$.

$$P = 2r(1 + 2 \sin(\theta))$$

Worked Example 3 - sector with internal triangle, perimeter expression

In the diagram, sector ABC has centre A , radius r and angle θ . D is the foot of the perpendicular from B to AC .

Express perimeter of triangle BCD in terms of r and θ .



1. In right triangle ABD :
 $AD = r \cos \theta$, $BD = r \sin \theta$.
2. $DC = AC - AD = r - r \cos \theta = r(1 - \cos \theta)$.
3. Straight chord $BC = 2r \sin\left(\frac{\theta}{2}\right)$.
4. Perimeter $BCD = r \sin \theta + r(1 - \cos \theta) + 2r \sin\left(\frac{\theta}{2}\right)$.

$$P = r \left(\sin(\theta) + 1 - \cos(\theta) + 2 \sin\left(\frac{\theta}{2}\right) \right)$$

Worked Examples - Area in r and θ

Two examples expressing area symbolically.

Worked Example 4 - shaded region in isosceles triangle minus sector

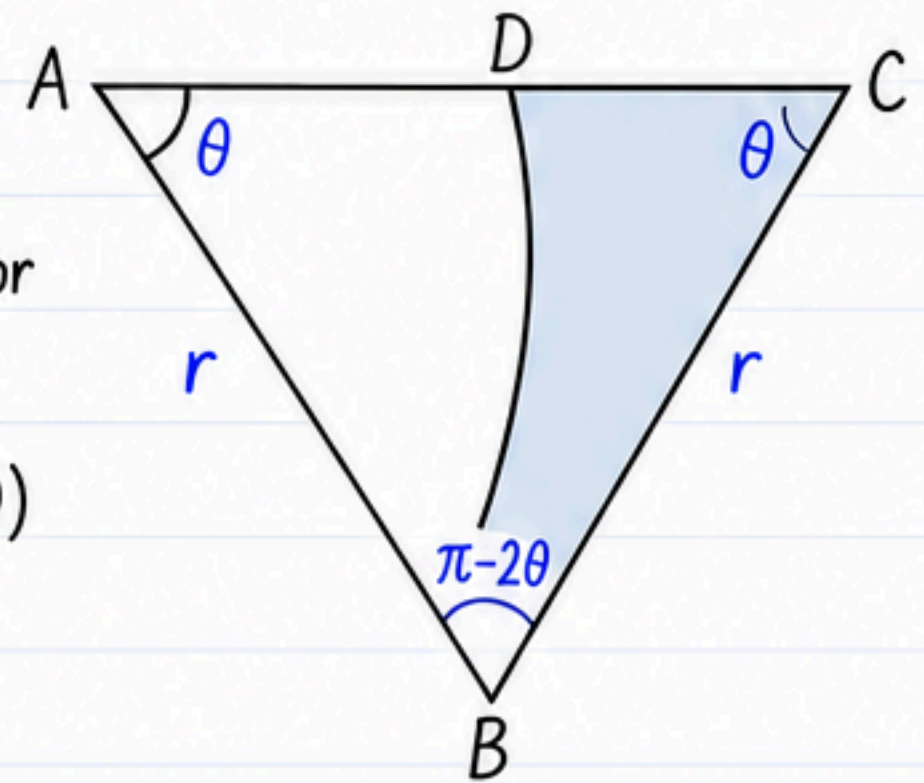
In the diagram, triangle ABC is isosceles with

$AB = BC = r$ and angle $BAC = \theta$.

Sector ABD has centre A , radius r and

angle θ . Assume $0 < \theta \leq \pi/3$, so the sector of radius r lies inside the triangle.

Express shaded area (triangle ABC minus sector ABD) in terms of r and θ .



1. angle $BCA = \theta$. angle $ABC = \pi - 2\theta$.

2. Triangle ABC area $= \left(\frac{1}{2}\right)(r)(r) \sin(\pi - 2\theta)$
 $= \frac{1}{2} r^2 \sin(2\theta)$.

3. Sector ABD area $= \frac{1}{2} r^2 \theta$.

4. Shaded area $= \frac{1}{2} r^2 \sin(2\theta) - \frac{1}{2} r^2 \theta$
 $= \frac{1}{2} r^2 (\sin(2\theta) - \theta)$.

$$\text{Shaded area} = \frac{1}{2} r^2 (\sin(2\theta) - \theta)$$

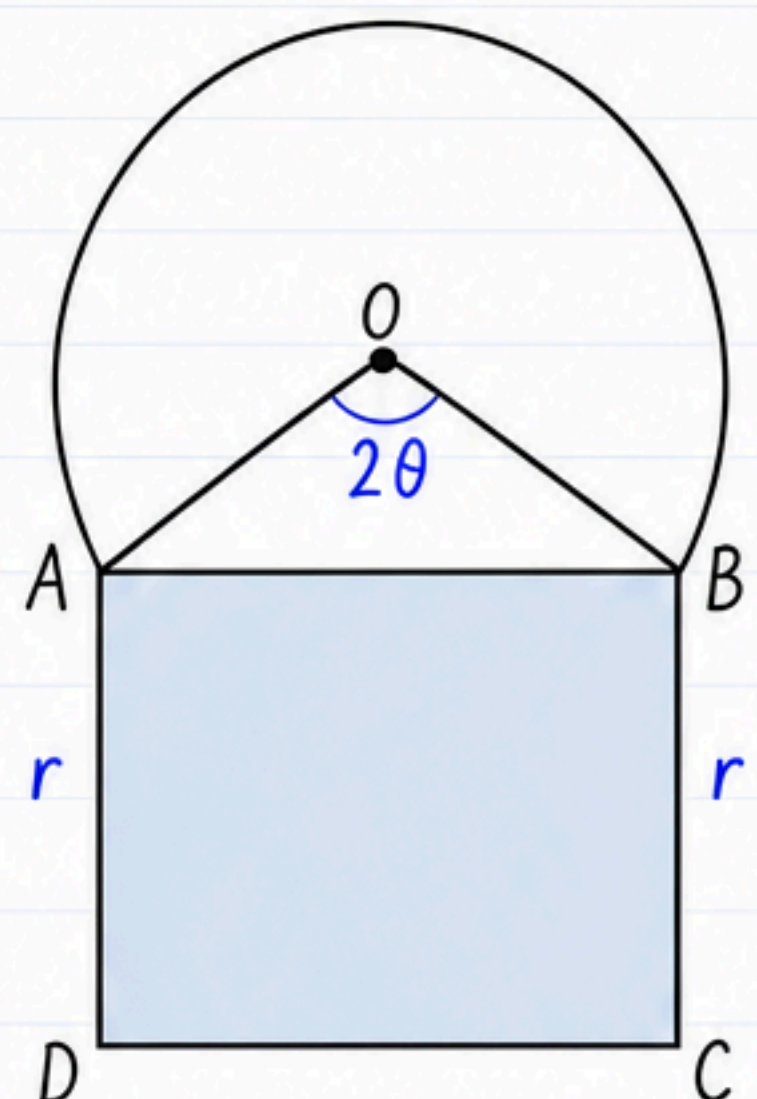
Worked Example 5 - shaded rectangle area

The diagram shows rectangle $ABCD$ attached below chord AB of a circle with centre O and radius r . $AD = r$ and angle $AOB = 2\theta$.

Find the area of the shaded rectangle.

1. Chord $AB = 2r \sin(\theta)$.

2. Rectangle area $= AB \times AD$
 $= (2r \sin(\theta))(r)$
 $= 2r^2 \sin(\theta)$.



$$\text{Shaded area} = 2r^2 \sin(\theta)$$

Worked Examples – Cosine Rule with Symbolic r

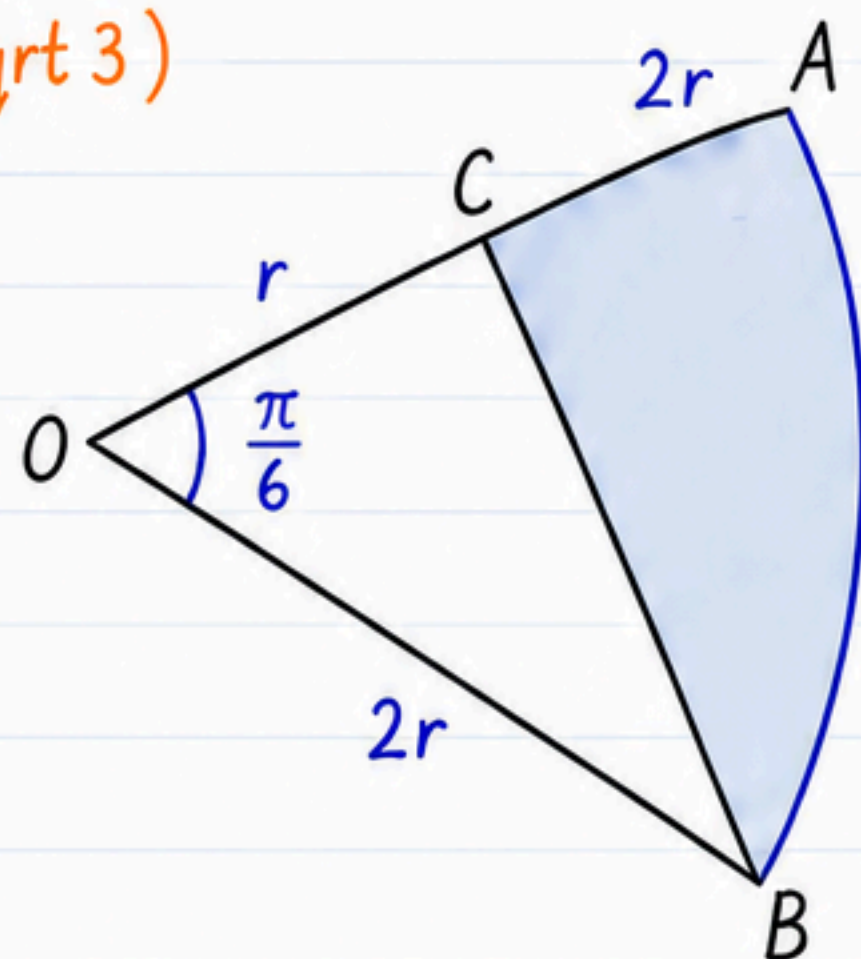
Two examples where cos rule produces a sqrt expression in r .

Worked Example 6 – $BC = r \sqrt{5 - 2\sqrt{3}}$

In the diagram, OAB is a sector of a circle with centre O and radius $2r$.

C is the midpoint of OA , so $OC = r$.

Show that $BC = r \sqrt{5 - 2\sqrt{3}}$.



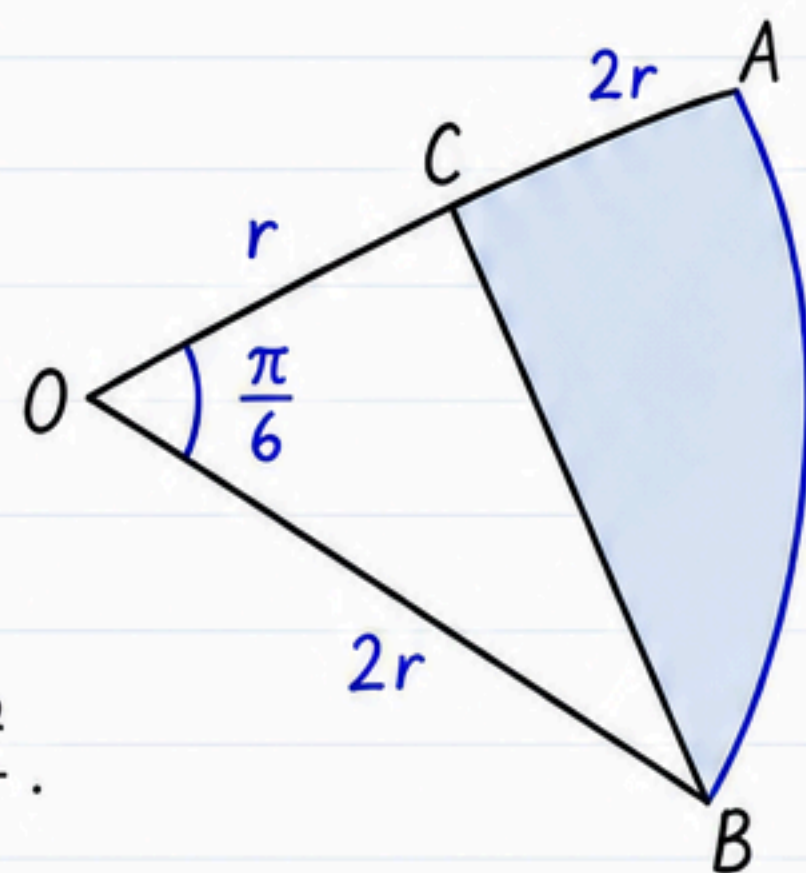
1. In triangle OBC : $OB = 2r$, $OC = r$, included angle $BOC = \frac{\pi}{6}$.
2. Cos rule: $BC^2 = OB^2 + OC^2 - 2 OB OC \cos\left(\frac{\pi}{6}\right)$
 $= 4r^2 + r^2 - 2(2r)(r)\left(\frac{\sqrt{3}}{2}\right).$
3. $BC^2 = 5r^2 - 2r^2\sqrt{3} = r^2(5 - 2\sqrt{3}).$
4. $BC = r \sqrt{5 - 2\sqrt{3}}$. AG.

$$BC = r \sqrt{5 - 2\sqrt{3}}$$

Worked Example 7 – area in r

In the same diagram as above, find shaded area in terms of r .

1. Sector OAB area $= \left(\frac{1}{2}\right)(2r)^2\left(\frac{\pi}{6}\right) = \frac{\pi r^2}{3}.$
2. Triangle OBC area $= \left(\frac{1}{2}\right)(r)(2r) \sin\left(\frac{\pi}{6}\right) = \frac{r^2}{2}.$
3. Shaded = sector - triangle
 $= \frac{\pi r^2}{3} - \frac{r^2}{2} = r^2\left(\frac{\pi}{3} - \frac{1}{2}\right) = r^2 \frac{(2\pi - 3)}{6}.$



$$\text{Shaded area} = r^2 \frac{(2\pi - 3)}{6}$$

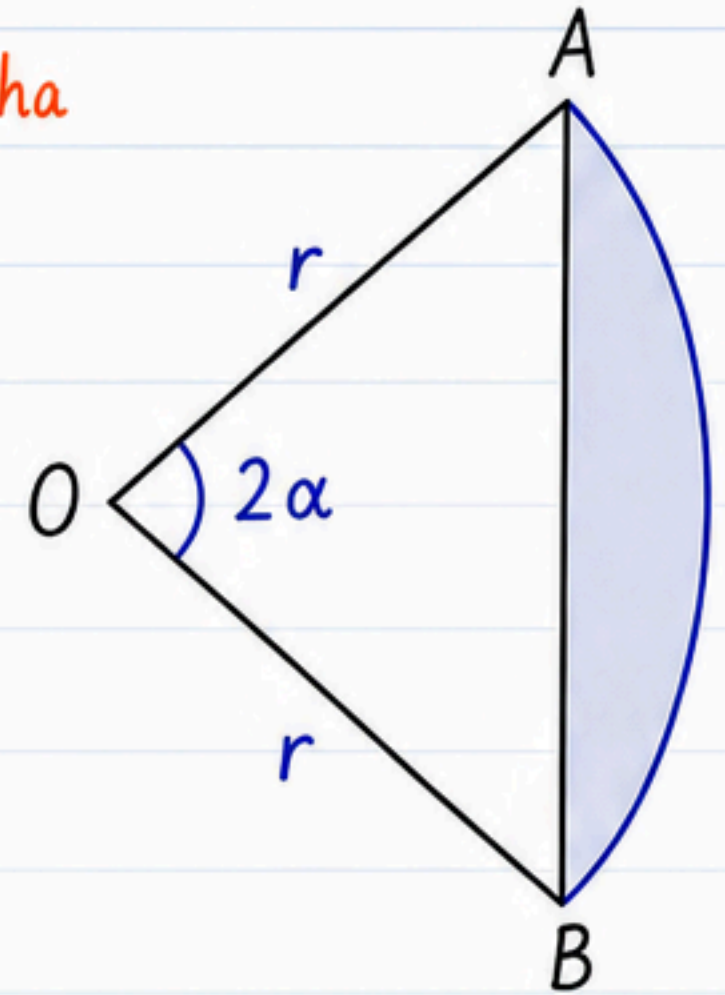
exact form with r^2 factored – matches Cambridge style for symbolic answers.

Worked Examples – Show That with Trig Identities

Two examples where the target uses $\sin(2\theta)$ or $\cos(2\theta)$ – apply identities.

Worked Example 8 – segment area in r and α

In the diagram, sector OAB has centre O , radius r and angle 2α . The segment is the region between chord AB and arc AB . Show segment area is $r^2 \left(\alpha - \frac{1}{2} \sin(2\alpha) \right)$.



1. Sector area = $\left(\frac{1}{2} \right) r^2 (2\alpha) = r^2 \alpha$.
2. Triangle OAB area = $\left(\frac{1}{2} \right) r^2 \sin(2\alpha)$.
3. Segment area = sector – triangle
 $= r^2 \alpha - \left(\frac{1}{2} \right) r^2 \sin(2\alpha)$
 $= r^2 \left(\alpha - \left(\frac{1}{2} \right) \sin(2\alpha) \right)$. AG.

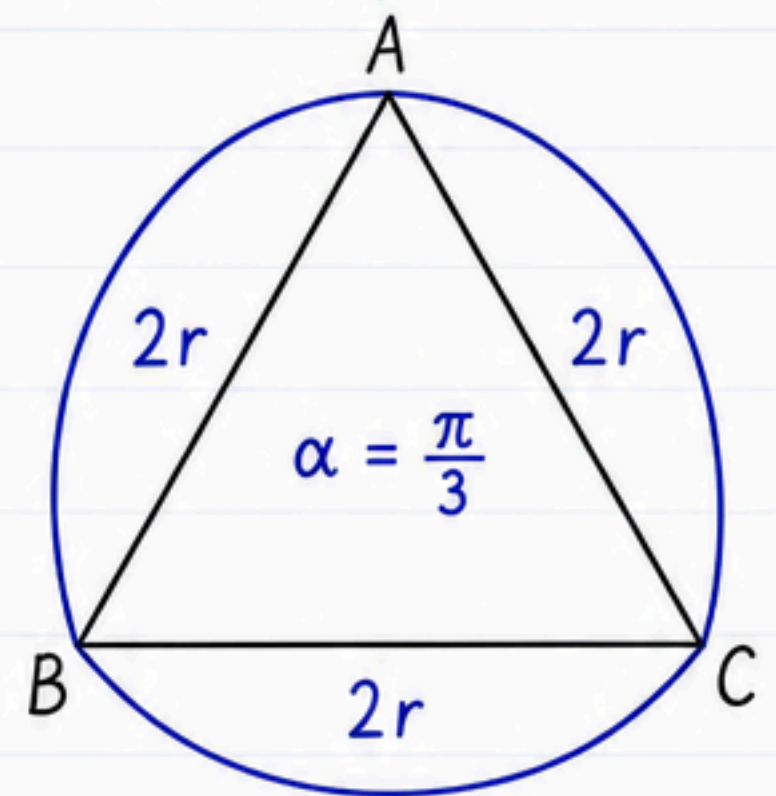
$$\text{Segment area} = r^2 \left(\alpha - \left(\frac{1}{2} \right) \sin(2\alpha) \right)$$

Worked Example 9 – equilateral plate with arcs in r and α

Equilateral triangle ABC has side length $2r$.

Arcs are centred at the vertices, each radius $2r$. Given $\alpha = \frac{\pi}{3}$.

Find exact area of the shape in terms of r .



1. Triangle ABC area = $\left(\frac{1}{2} \right) (2r)^2 \sin\left(\frac{\pi}{3}\right) = r^2 \sqrt{3}$.
2. Each external segment area = $\left(\frac{1}{2} \right) (2r)^2 \left(\frac{\pi}{3} - \sin\left(\frac{\pi}{3}\right) \right)$
 $= 2r^2 \left(\frac{\pi}{3} - \frac{\sqrt{3}}{2} \right)$.
3. Three segments total = $6r^2 \left(\frac{\pi}{3} - \frac{\sqrt{3}}{2} \right)$
 $= 2\pi r^2 - 3r^2 \sqrt{3}$.
4. Shape area = triangle + 3 segments
 $= r^2 \sqrt{3} + (2\pi r^2 - 3r^2 \sqrt{3})$
 $= 2\pi r^2 - 2r^2 \sqrt{3}$
 $= 2r^2 (\pi - \sqrt{3})$.

$$\text{Shape area} = 2r^2 (\pi - \sqrt{3})$$

The arcs bow outward, so add the external segments to the triangle.

Watch Out

Three slips that lose easy marks on symbolic and show-that questions.

Substituting numerical values mid-way



Don't do this.

"in terms of r and θ ":

let $r = 1$, plug in to test -
then forget to put r back.



Do this.

keep r symbolic ALL the way
to the final line. Substitute
only at the end if the question
gives a numerical r .

Leaving $\sin(2\theta)$ when target uses $2 \sin \theta \cos \theta$



Don't do this.

final line:

$$\text{shaded area} = r^2 \left(\alpha - \frac{\sin(2\alpha)}{2} \right)$$

(target uses $\sin \alpha \cos \alpha$ form)



Do this.

apply $\sin(2\alpha) = 2 \sin \alpha \cos \alpha$:
$$= r^2 (\alpha - \sin \alpha \cos \alpha)$$

matching the target.

Skipping algebra steps in AG questions



Don't do this.

"by isosceles,
 $OA = 2 \times \cos \theta$.
Done."

(no perpendicular drop,
no derivation shown)



Do this.

drop perpendicular from P to OA
at midpoint M .
In right triangle OPM :
 $OM = OP \cos \theta = x \cos \theta$.
So $OA = 2 OM = 2 \times \cos \theta$.

Make the equivalence to the target clear. For example,
 $\cos(\theta) \times 2x$ and $2x \cos(\theta)$ are the same product,
so either factor order is valid.

Try It Yourself

Cover the green box, work it out fully, then check. Three problems – each is heavy. Take your time.

1. Find the area of a sector of a circle of radius r and angle θ in terms of r and θ .

$$A_{\text{sector}} = \left(\frac{1}{2}\right) r^2 \theta.$$

2. In triangle OAP , $OP = AP = x$ and angle $AOP = \theta$. Show that $OA = 2x \cos \theta$.

Triangle OAP is isosceles ($OP = AP$),
so angle $OAP = \text{angle } AOP = \theta$.
Drop perpendicular from P to OA at midpoint M .
In right triangle OPM :

$$OM = OP \cos \theta = x \cos \theta.$$
$$OA = 2 OM = 2x \cos \theta. \quad \text{AG.}$$

3. A sector has radius r and angle 2θ . Show that the segment area equals $r^2 \left(\theta - \left(\frac{1}{2}\right) \sin(2\theta)\right)$.

$$\text{Sector area} = \left(\frac{1}{2}\right) r^2 (2\theta) = r^2 \theta.$$

$$\text{Triangle } OAB \text{ area} = \left(\frac{1}{2}\right) r^2 \sin(2\theta).$$

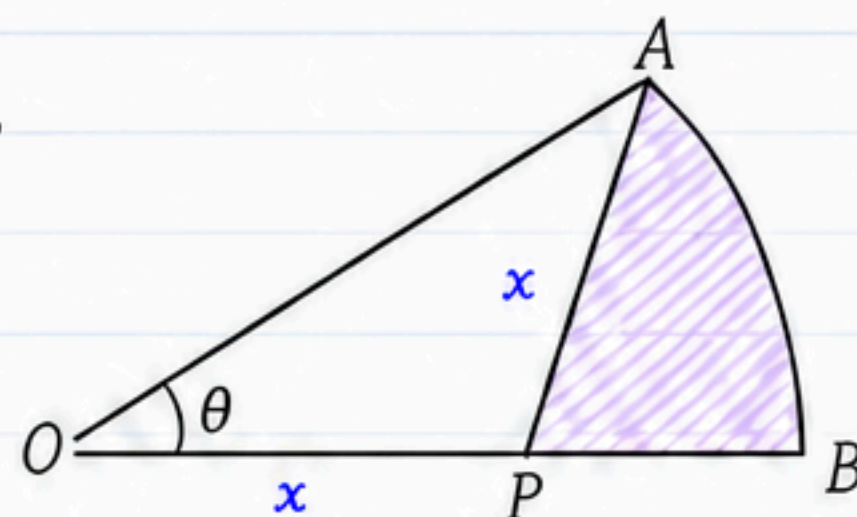
$$\begin{aligned} \text{Segment} &= \text{sector} - \text{triangle} = r^2 \theta - \left(\frac{1}{2}\right) r^2 \sin(2\theta) \\ &= r^2 \left(\theta - \left(\frac{1}{2}\right) \sin(2\theta)\right). \quad \text{AG.} \end{aligned}$$

Exam-Style Practice

1. Sector OAB has centre O . P lies on OB , $OP = AP = x$, and angle $AOB = \theta$.

(a) Show $\text{arc } AB = 2x\theta \cos \theta$.

(b) Find the shaded area APB in terms of x and θ . [6]



Triangle OAP is isosceles ($OP = AP = x$), so angle $OAP = \theta$.

Drop perpendicular PM to OA ; M is the midpoint of OA .

In right triangle OPM , $OM = OP \cos \theta = x \cos \theta$.

Hence $OA = 2OM = 2x \cos \theta$.

Arc $AB = OA \times \theta = (2x \cos \theta) \theta = 2x\theta \cos \theta$. AG.

Area of sector $OAB = \left(\frac{1}{2}\right)(OA)^2 \theta = \left(\frac{1}{2}\right)(2x \cos \theta)^2 \theta = 2x^2 \theta \cos^2 \theta$.

Area of triangle $OAP = \left(\frac{1}{2}\right)(OP)(OA) \sin \theta = \left(\frac{1}{2}\right)(x)(2x \cos \theta) \sin \theta$
 $= x^2 \sin \theta \cos \theta$.

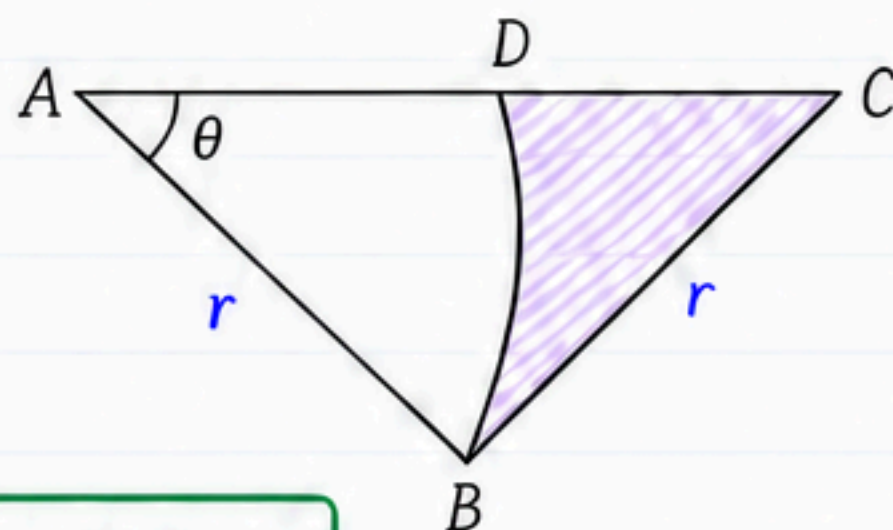
Shaded area $APB = \text{sector } OAB - \text{triangle } OAP$

$$= 2x^2 \theta \cos^2 \theta - x^2 \sin \theta \cos \theta$$

$$= x^2 \cos \theta [2\theta \cos \theta - \sin \theta].$$

2. In an isosceles triangle ABC with $AB = BC = r$ and angle $BAC = \theta$, point D lies on AC . Sector ABD has centre A and radius r .

Express the shaded area in terms of r and θ . [4]



Area of triangle $ABC = \left(\frac{1}{2}\right)r^2 \sin(\pi - 2\theta) = \left(\frac{1}{2}\right)r^2 \sin(2\theta)$.

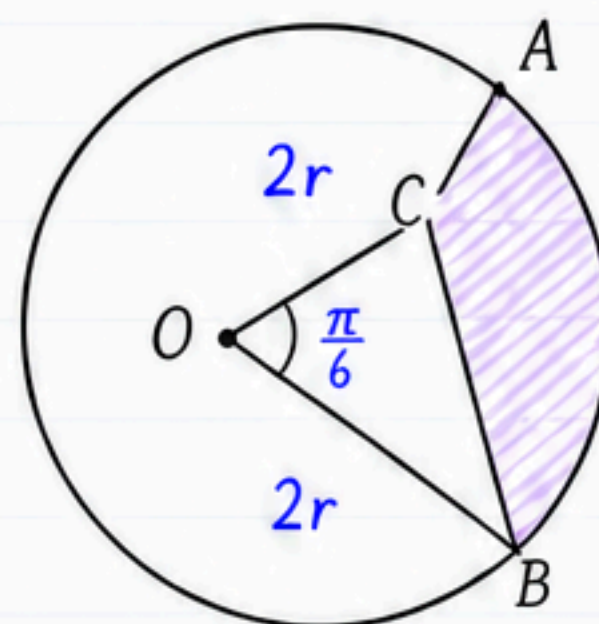
Area of sector $ABD = \left(\frac{1}{2}\right)r^2 \theta$.

Shaded area (triangle DBC) = area of triangle ABC - area of sector ABD
 $= \left(\frac{1}{2}\right)r^2 \sin(2\theta) - \left(\frac{1}{2}\right)r^2 \theta$
 $= \left(\frac{1}{2}\right)r^2 [\sin(2\theta) - \theta]$.

3. Sector OAB has centre O , radius $2r$, and angle $\pi/6$. C is the midpoint of OA .

(a) Show that $BC = r\sqrt{5 - 2\sqrt{3}}$.

(b) Find the perimeter of the shaded region (arc AB + BC + CA) in terms of r . [5]



In triangle OBC : $OB = 2r$, $OC = r$, and angle $BOC = \pi/6$.

By the cosine rule,

$$BC^2 = (2r)^2 + r^2 - 2(2r)(r) \cos\left(\frac{\pi}{6}\right)$$

$$= 4r^2 + r^2 - 4r^2 \left(\frac{\sqrt{3}}{2}\right) = r^2(5 - 2\sqrt{3}).$$

Hence $BC = r\sqrt{5 - 2\sqrt{3}}$. AG.

Arc $AB = (2r)(\pi/6) = \pi r/3$.

Perimeter = arc AB + BC + CA

$$= \pi r/3 + r\sqrt{5 - 2\sqrt{3}} + r$$

$$= \pi r/3 + r + r\sqrt{5 - 2\sqrt{3}}.$$